

1. FUNCTION

```
mysql> use 22a;  
Database changed  
mysql> DELIMITER //  
mysql>  
mysql> CREATE FUNCTION GetSquare(num INT)  
-> RETURNS INT  
-> DETERMINISTIC  
-> BEGIN  
->     RETURN num * num;  
-> END //
```

Query OK, 0 rows affected (0.06 sec)

```
mysql>  
mysql> DELIMITER ;  
mysql>  
mysql> SELECT GetSquare(5);
```

GetSquare(5)
25

1 row in set (0.01 sec)

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION CheckNumber(num INT)
-> RETURNS VARCHAR(20)
-> DETERMINISTIC
-> BEGIN
->     IF num >= 0 THEN
->         RETURN 'Positive';
->     ELSE
->         RETURN 'Negative';
->     END IF;
-> END //
```

Query OK, 0 rows affected (0.01 sec)

```
mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT CheckNumber(10);
+-----+
| CheckNumber(10) |
+-----+
| Positive        |
+-----+
1 row in set (0.00 sec)
```

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION GetGrade(mark INT)
-> RETURNS VARCHAR(10)
-> DETERMINISTIC
-> BEGIN
->     IF mark >= 90 THEN
->         RETURN 'A';
->     ELSEIF mark >= 75 THEN
->         RETURN 'B';
->     ELSEIF mark >= 50 THEN
->         RETURN 'C';
->     ELSE
->         RETURN 'Fail';
->     END IF;
-> END //
```

Query OK, 0 rows affected (0.01 sec)

```
mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT GetGrade(85);
+-----+
| GetGrade(85) |
+-----+
| B            |
+-----+
1 row in set (0.00 sec)
```

```

mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION Grade(mark INT)
-> RETURNS VARCHAR(10)
-> DETERMINISTIC
-> BEGIN
->     DECLARE result VARCHAR(10);
->
->     CASE
->         WHEN mark >= 90 THEN
->             SET result = 'A';
->         WHEN mark >= 75 THEN
->             SET result = 'B';
->         WHEN mark >= 50 THEN
->             SET result = 'C';
->         ELSE
->             SET result = 'FAIL';
->     END CASE;
->
->     RETURN result;
-> END //

```

Query OK, 0 rows affected (0.02 sec)

```

mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT Grade(85);
+-----+
| Grade(85) |
+-----+
| B         |
+-----+
1 row in set (0.00 sec)

```

```

mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION SumLoop(n INT)
-> RETURNS INT
-> DETERMINISTIC
-> BEGIN
->     DECLARE i INT DEFAULT 1;
->     DECLARE total INT DEFAULT 0;
->
->     loop1: LOOP
->         SET total = total + i;
->         SET i = i + 1;
->
->         IF i > n THEN
->             LEAVE loop1;
->         END IF;
->     END LOOP;
->
->     RETURN total;
-> END //

```

Query OK, 0 rows affected (0.01 sec)

```

mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT SumLoop(5);
+-----+
| SumLoop(5) |
+-----+
|          15 |
+-----+
1 row in set (0.00 sec)

```

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION SumWhile(n INT)
-> RETURNS INT
-> DETERMINISTIC
-> BEGIN
->     DECLARE i INT DEFAULT 1;
->     DECLARE total INT DEFAULT 0;
->
->     WHILE i <= n DO
->         SET total = total + i;
->         SET i = i + 1;
->     END WHILE;
->
->     RETURN total;
-> END //
Query OK, 0 rows affected (0.02 sec)
```

```
mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT SumWhile(5);
+-----+
| SumWhile(5) |
+-----+
|          15 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION SumRepeat(n INT)
-> RETURNS INT
-> DETERMINISTIC
-> BEGIN
->     DECLARE i INT DEFAULT 1;
->     DECLARE total INT DEFAULT 0;
->
->     REPEAT
->         SET total = total + i;
->         SET i = i + 1;
->     UNTIL i > n
->     END REPEAT;
->
->     RETURN total;
-> END //
Query OK, 0 rows affected (0.02 sec)
```

```
mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT SumRepeat(5);
+-----+
| SumRepeat(5) |
+-----+
|          15 |
+-----+
1 row in set (0.00 sec)
```